



Harnessing AI capabilities and green entrepreneurial orientation for sustainable SME performance using SEM analysis approach

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ABSTRACT

The growing focus on sustainability and technological innovation has encouraged small and medium-sized enterprises (SMEs) to adopt artificial intelligence (AI) capabilities and a green entrepreneurial orientation as central drivers of sustainable performance. This research investigates the contribution of AI capabilities in promoting green innovations and creativity in SMEs, which collectively lead to long-term sustainability. The study is based on the Resource-Based View (RBV) theory, which offers a theoretical framework for investigating how AI-based competencies and green entrepreneurial strategies can promote SME performance. The research model comprises infrastructure, business integration, and a proactive attitude influencing AI capabilities, affecting SME creativity and green innovations. Moreover, green risk-taking, innovativeness, and proactiveness significantly contribute to green entrepreneurial orientation, affecting SME creativity and green innovations, ultimately resulting in sustainable performance. A quantitative research design was applied, utilizing survey data gathered from SME managers and business owners functioning in the manufacturing and service sectors in Saudi Arabia. 250 SMEs were examined using “Structural Equation Modeling (SEM)” analysis to validate the proposed hypotheses and evaluate these relationships. The results indicate that AI capabilities significantly impact SME creativity and green innovations. Moreover, green entrepreneurial orientation positively influences SME creativity and green innovations, which in turn facilitate sustainable performance. This research identifies the importance of SMEs allocating resources to AI infrastructure, proactive business strategies, and entrepreneurial risk-taking to foster green innovation and sustainability. Moreover, policymakers must facilitate AI-based green initiatives by implementing incentives and regulatory policies to promote sustainable development. By integrating AI capabilities and green entrepreneurship, SMEs can achieve a competitive advantage while achieving world sustainability goals.

1. Introduction

Small and Medium Enterprises (SMEs) are the backbone of the global economy, playing an important role in job creation, innovation, and economic change (Alshebami, 2024; Al-Zubaidi et al., 2024). Its contribution is particularly significant in emerging economies, supporting industrial development and enhancing economic resilience. In Saudi Arabia, SMEs are the top priority of the Vision 2030 initiative, which aims to transition the economy from oil dependency to knowledge-based (Aldhobaib, 2025). The sustainability agenda has gained prominence within this transformation, compelling the adoption of environmentally responsible practices (Aldhobaib, 2025; Aboudayah,

2023). Green innovation is developing and applying eco-friendly processes, products, and business models. It has become a fundamental component of sustainable business strategies (Patwary et al., 2024; Le, 2022). While regulatory policies, user preferences, and commercial social responsibility drive green innovation adoption, its successful implementation largely depends on technological advancements and managerial capabilities. Saudi Arabia's Vision 2030 highlights the serious role of SMEs in achieving economic expansion and reducing dependence on oil revenues. The government has introduced multiple initiatives to support SME growth, including financial incentives, digital transformation programs, and regulatory reforms (Aldhobaib, 2025). However, despite these efforts, SMEs in Saudi Arabia face significant

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